

PH 716 Applied Survival Analysis

Part 9: Competing risks

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What are competing risks?

- In many real-world studies, a subject can experience one of **several different types of events**. In practice, we only observe the first event, and follow-up stops afterward. As a result, once one event is observed, the other event types are no longer observable for that subject.

Examples of competing risks in medical studies

- A person may die from: natural causes, accidental death, homicide, suicide, etc.
- A patient may develop: heart disease, diabetes, kidney diseases, etc.
- If one event is observed, the follow-up will be terminated.

Data structure for competing risks

- i : subject index, $i = 1, \dots, n$
- $\tilde{T}_i = \min(T_i, C_i)$: observed time for subject i
 - T_i : authentic time to first event for subject i
 - C_i : censoring time for subject i
- Δ_i : the type of the first observed event for subject i
 - $\Delta_i = k$, $k = 1, \dots, K$: $T_i = \tilde{T}_i$ and the first event is of type k
 - $\Delta_i = 0$: $T_i = C_i$, i.e., the subject is censored before any event occurs
- $x_{i1}, \dots, x_{ip}$: values of covariates for subject i

Recall the hazard function

- Limited to continuous T_i
- Hazard function

$$\lambda_{T_i}(t) = \lim_{\delta \rightarrow 0^+} \frac{\Pr(t \leq T_i < t + \delta \mid T_i \geq t)}{\delta}$$

- The instantaneous risk of experiencing one event at time t , assuming the subject has survived up to t
 - NOT accounting for competing risks
- Why event type matters? If we ignore event types and treat all events as the same, we may miss important patterns.
 - Example: A drug effect might strongly extend the cancer-free survival but not survival from other causes.

Cause-specific hazard

- Instead of one hazard shared by all event types, we now have one per type:

$$\lambda_{T_i}^{(k)}(t) = \lim_{\delta \rightarrow 0^+} \frac{\Pr(t \leq T_i < t + \delta, \Delta_i = k \mid T_i \geq t)}{\delta}, \quad k = 1, \dots, K$$

- Interpretation: at time t , the instantaneous risk of event type k , given the subject has not had any event yet
- Total instantaneous risk = sum of instantaneous risks for each event type: $\lambda_{T_i}(t) = \sum_{k=1}^K \lambda_{T_i}^{(k)}(t)$
- Cumulative cause-specific hazard $\Lambda_{T_i}^{(k)}(t) = \int_0^t \lambda_{T_i}^{(k)}(u) du$
 - Total cumulative risk = sum of cumulative risks for each event type: $\Lambda_{T_i}(t) = \sum_{k=1}^K \Lambda_{T_i}^{(k)}(t)$

Cumulative incidence function (CIF, aka sub-distribution function)

- CIF: $F_{T_i}^{(k)}(t) = \Pr(T_i \leq t, \Delta_i = k)$
 - Interpretation: the probability that the first observed event is of type k and occurs by time t
 - CDF = sum of CIFs for each event type: $F_{T_i}(t) = \sum_{k=1}^K F_{T_i}^{(k)}(t) \Rightarrow S_{T_i}(t) = 1 - \sum_{k=1}^K F_{T_i}^{(k)}(t)$
 - NOT a cumulative distribution function (CDF) because $F_{T_i}^{(k)}(\infty) < F_{T_i}(\infty) = 1$
- Theoretical notes
 - $\lambda_{T_i}^{(k)}(t) = \frac{dF_{T_i}^{(k)}(t)/dt}{S_{T_i}(t)}$
 - * Proof: $\lambda_{T_i}^{(k)}(t) = \lim_{\delta \rightarrow 0^+} \frac{\Pr(t \leq T_i < t + \delta, \Delta_i = k, T_i \geq t)}{\delta \Pr(T_i \geq t)} = \lim_{\delta \rightarrow 0^+} \frac{\Pr(t \leq T_i < t + \delta, \Delta_i = k)}{\delta S_{T_i}(t)} = \frac{dF_{T_i}^{(k)}(t)/dt}{S_{T_i}(t)}$
 - $F_{T_i}^{(k)}(t) = \int_0^t \lambda_{T_i}^{(k)}(u) S_{T_i}(u) du$ (backing up KM estimators of CIFs and overall survival)

When covariates are not considered: naive KM estimator

- Assuming that
 - T_i iid across i , i.e., $T_i \stackrel{\text{iid}}{\sim} T$
 - T_i independent of C_i
 - Times to different events are independent (typically violated in medical cases)
 - * Implying that at each time point the hazard of each event is the same for subjects at risk as for subjects that have experienced other competing events by that time
- Implementation
 - Given k , take the event type k as the event of interest with other types considered as censored
 - Apply KM estimator to the resulting binary setting
- Theoretical notes
 - Naive KM estimator gives cause-specific survival by $\prod_{j:t_j \leq t} \{1 - \hat{\lambda}_T^{(k)}(t_j)\}$
 - * $0 = t_0 < t_1 < \dots < t_J$: unique failure times
 - * $\hat{\lambda}_T^{(k)}(t_j) = d_{kj}/r_j$: an estimate of the cause-specific hazard function
 - d_{kj} : # of event k that happened exactly at time t_j
 - r_j : # of individuals at risk up to time t_j
- Underestimating the survival probability (i.e., overestimating the failure probability)
 - Taking other types as censoring mistakenly assumes people who had other events could still experience the event of interest later.
 - The bias inflated when the competition when the hazards of competing events are larger.

Ex. 9.1 High risk population in `asaur::prostateSurvival`

- Dataset `asaur::prostateSurvival` involves covariates as below.
 - **grade**: a factor with levels `mod` (moderately differentiated) and `poor` (poorly differentiated)
 - **stage**: a factor with levels `T1ab` (Stage T1, clinically diagnosed), `T1c` (Stage T1, diagnosed via a PSA test), and `T2` (Stage T2)
 - **ageGroup**: a factor with levels `66-69`, `70-74`, `75-79`, & `80+`

- **survTime**: the survival time from diagnosis to death (from prostate cancer or other causes) or last date known alive
- **status**: a censoring variable, 0 (censored), 1 (death from prostate cancer), and 2 (death from other causes)
- Use naive KM estimators to give the survival curve of high risk population (i.e., **grade="poor"**, **stage="T2"** & **ageGroup="80+"**) for each event type.

```
options(digits=4)
library(asaaur)
library(survival)
sapply(asaaur::prostateSurvival, class)
data.ex91 = asaaur::prostateSurvival[
  asaaur::prostateSurvival$grade == "poor" &
  asaaur::prostateSurvival$stage == "T2" &
  asaaur::prostateSurvival$ageGroup == "80+"
,
]
km.prost.naive = survfit(
  Surv(survTime, event=(data.ex91$status==1)) ~ 1,
  data=data.ex91
)
km.other.naive = survfit(
  Surv(survTime, event=(data.ex91$status==2)) ~ 1,
  data=data.ex91
)
plot(
  km.prost.naive$surv ~ km.prost.naive$time, type="s", ylim=c(0,1), lwd=2, col="blue",
  xlab="Months from prostate cancer diagnosis",
  ylab='Estimated survival probability',
)
lines(km.other.naive$surv ~ km.other.naive$time, type="s", col="green", lwd=2)
legend(
  "topright",
  c(
    "Prostate",
    "Other"
  ),
  col=c('blue','green'), lwd=2
)
```

When covariates are not considered: KM estimator of CIF

- Assuming that
 - T_i iid across i , i.e., $T_i \stackrel{\text{iid}}{\sim} T$
 - T_i independent of C_i
- Implementation
 - Instead of naive KM: estimate overall survival and then combine it with cause-specific hazards
 - Estimate overall survival $S_T(t)$ by $\hat{S}_{T,KM}(t) = \prod_{j:t_j \leq t} \{1 - \sum_{k=1}^K \hat{\lambda}_T^{(k)}(t_j)\}$
 - * $0 = t_0 < t_1 < \dots < t_J$: unique failure times
 - * $\hat{\lambda}_T^{(k)}(t_j) = d_{kj}/r_j$: an estimate of the cause-specific hazard function
 - d_{kj} : # of event k that happened exactly at time t_j
 - r_j : # of individuals at risk up to time t_j
 - Estimate CIF $F_T^{(k)}(t)$ by $\hat{F}_{T,KM}^{(k)}(t) = \sum_{j:t_j \leq t} \hat{\lambda}_T^{(k)}(t_j) \hat{S}_{T,KM}(t_j - 1)$

Revisit Ex. 9.1

Use the KM estimator of CIF to give the survival curve of high risk population

```
options(digits=4)
library(asaur)
library(survival)
library(mstate)
sapply(asaur::prostateSurvival, class)
data.ex91 = asaur::prostateSurvival[
  asaur::prostateSurvival$grade == "poor" &
  asaur::prostateSurvival$stage == "T2" &
  asaur::prostateSurvival$ageGroup == "80+"
,
]
km.cif = Cuminc(
  time = data.ex91$survTime,
  status = data.ex91$status
)
km.cif

# Plot of CIFs and the overall survival function
plot(
  km.cif$CI.1 ~ km.cif$time, type="s", ylim=c(0,1), lwd=2, col="blue",
  xlab="Months from prostate cancer diagnosis",
  ylab="Probability"
)
lines(km.cif$CI.2 ~ km.cif$time, type="s", lwd=2, col="green")
lines(km.cif$Surv ~ km.cif$time, type="s", lwd=2, col="red")
legend(
  "topright",
  c(
    "CIF (prostate)",
    "CIF (other)",
    "Overall survival"
  ),
  col=c('blue','green','red'), lwd=2
)

# Stacked plot
library(ggplot2)
cuminc_data = as.data.frame(km.cif[, c('time','Surv','CI.1','CI.2')])
cuminc_data = tidyr::pivot_longer(
  cuminc_data, cols = -time, names_to = "Types", values_to = "estimate")
ggplot(data = cuminc_data, aes(x = as.numeric(time), y = estimate, fill = Types)) +
  geom_area(alpha = 0.6) +
  labs(x = "Months from prostate cancer diagnosis", y = "Probability") +
  theme_minimal()
```

Modeling effects of covariates: cause-specific Cox PH model

- Assuming that
 - T_i independent across i given covariates
 - The independent and non-informative censoring
 - Cause-specific proportional hazards

- * $\lambda_{T_i}^{(k)}(t) = \lambda_0^{(k)}(t) \exp(\sum_{j=1}^p x_{ij} \beta_j^{(k)})$
 - $\lambda_0^{(k)}(t)$: baseline cause-specific hazard of event k
 - $\beta_1^{(k)}, \dots, \beta_p^{(k)}$: covariate effects varying from one event to another
- * OR $\lambda_{T_i}^{(k)}(t) = \lambda_0^{(k)}(t) \exp(\sum_{j=1}^p x_{ij} \beta_j)$, i.e., β_j shared by all the K events
- Estimation procedure
 - For $\lambda_{T_i}^{(k)}(t) = \lambda_0^{(k)}(t) \exp(\sum_{j=1}^p x_{ij} \beta_j^{(k)})$
 - * Specify one event of interest and fit a Cox PH model with the remaining $K - 1$ events treated as censoring
 - * Repeat the above step and obtain K Cox PH models
 - For $\lambda_{T_i}^{(k)}(t) = \lambda_0^{(k)}(t) \exp(\sum_{j=1}^p x_{ij} \beta_j)$
 - * First reshape the data frame in the “long format” by encoding the each row with K rows
 - * Then fit a Cox PH model stratified by the encoded event label in the long format
- When $\hat{\lambda}_{T_i}^{(k)}(t)$ is ready
 - $\hat{S}_{T_i}(t) = \exp\{-\sum_{k=1}^K \int_0^t \hat{\lambda}_{T_i}^{(k)}(u) du\}$
 - $\hat{F}_{T_i}^{(k)}(t) = \int_0^t \hat{\lambda}_{T_i}^{(k)}(u) \hat{S}_{T_i}(u) du$
- Pros and cons
 - Easy to implement
 - Straightforward interpretation of regression coefficients in terms of hazard ratio BUT inconvenient to interpret coefficients in terms of contributions to CIFs
 - Bias induced since we treat other events as censoring

Ex. 9.2 Patients at “T2”-stage in `asaur::prostateSurvival`

- Consider patients with `stage="T2"`.

```
options(digits=4)
library(asaur)
library(survival)
sapply(asaur::prostateSurvival, class)
data.ex92 = asaur::prostateSurvival[
  asaur::prostateSurvival$stage == "T2"
,
]

# Regression on cause-specific hazards
data.ex92$status.1 = (data.ex92$status==1)
data.ex92$status.2 = (data.ex92$status==2)
cph.prost = coxph(
  Surv(survTime, status.1)~grade + ageGroup,
  data = data.ex92
)
summary(cph.prost)
cph.other = coxph(
  Surv(survTime, status.2)~grade + ageGroup,
  data = data.ex92
)
summary(cph.other)

# Regression on cause-specific hazards with shared coefficients
## Reshape the data into the long format
data.ex92.long = NULL
K = length(unique(data.ex92$status))-1
for (i in 1:nrow(data.ex92)){
```

```

data.curr = data.ex92[rep(i, times=K),]
data.curr$event = c('prostate', 'other')
data.curr$status.long=rep(0,K-1)
if(data.ex92$status[i]>=1) {
  data.curr$status.long[which(data.curr$event==c('prostate', 'other')[data.ex92$status[i]])]=1
}
data.ex92.long = rbind(data.ex92.long, data.curr)
}
data.ex92.long = data.ex92.long[
  ,
  !(names(data.ex92.long) %in% c('status.1','status.2'))] # remove columns to avoid confusion
head(data.ex92)
head(data.ex92.long)

## Equivalency between two data frames
head(data.ex92)
head(data.ex92.long[data.ex92.long$event=='prostate',])
head(data.ex92.long[data.ex92.long$event=='other',])

## Cox PH model stratified by event
cph.strat = coxph(
  Surv(survTime, status.long)~grade + ageGroup+strata(event),
  data = data.ex92.long
)
summary(cph.strat)

```

Sub-distribution hazard function

$$\bar{\lambda}_{T_i}^{(k)}(t) = -\frac{d \ln\{1 - F_{T_i}^{(k)}(t)\}}{dt} = \frac{dF_{T_i}^{(k)}(t)/dt}{1 - F_{T_i}^{(k)}(t)}, \quad k = 1, \dots, K$$

- Interpretation: a quantity that determines how the probability of event type k grows over time, accounting for competing risks.
- NOT the cause-specific hazard function: $\bar{\lambda}_{T_i}^{(k)}(t) \leq \lambda_{T_i}^{(k)}(t)$
- Theoretical note: $F_{T_i}^{(k)}(t) = 1 - \exp\{-\int_0^t \bar{\lambda}_{T_i}^{(k)}(u)du\}$ (backing up the Fine-Gray model)

Modeling effects of covariates: Fine-Gray model

- Assuming that
 - T_i independent across i given covariates
 - The independent and non-informative censoring
 - $\bar{\lambda}_{T_i}^{(k)}(t) = \bar{\lambda}_0^{(k)}(t) \exp(\sum_{j=1}^p x_{ij}\beta_j^{(k)})$
 - * $\bar{\lambda}_0^{(k)}(t)$: baseline sub-distribution hazard of event k
 - * $\beta_1^{(k)}, \dots, \beta_p^{(k)}$: covariate effects potentially varying from one event to another
- When $\hat{\lambda}_{T_i}^{(k)}(t)$ is ready
 - $\hat{F}_{T_i}^{(k)}(t) = 1 - \exp\{-\int_0^t \hat{\lambda}_{T_i}^{(k)}(u)du\}$
- Pros and cons
 - Direct modeling of CIFs
 - Difficult to interpret regression coefficients in terms of hazard ratio, BUT may see the effect of covariates to CIFs directly
 - Might not satisfy the additive restriction $\sum_{k=1}^K \hat{F}_{T_i}^{(k)}(t) \leq 1$

Revisit Ex. 9.2

- Poorly differentiated patients (`grade=poor`) have higher risk for death from both prostate and other.
- Elder patients also have higher risk for the death from both conditions.

```
options(digits=4)
library(asauro)
data.ex92 = asauro::prostateSurvival[
  asauro::prostateSurvival$stage == "T2"
,
]
# Model fitting
cov1 = model.matrix(~ grade + ageGroup, data = data.ex92)[-1]
cph.subdisthz.prost = cmprsk::crr(
  ftime = data.ex92$survTime,
  fstatus = data.ex92$status,
  cov1 = cov1,
  failcode=1
)
summary(cph.subdisthz.prost)
cph.subdisthz.other = cmprsk::crr(
  ftime = data.ex92$survTime,
  fstatus = data.ex92$status,
  cov1 = cov1,
  failcode=2
)
summary(cph.subdisthz.other)

# Predicted CIFs
cov1_new = matrix( # Specifying subjects with grade=poor and age= 70-74
  c(1, 1, 0, 0),
  byrow = T,
  ncol = 4
)
colnames(cov1_new) = colnames(cov1); cov1_new
predict.prost = predict(
  cph.subdisthz.prost,
  cov1 = cov1_new
)
head(predict.prost)

predict.other = predict(
  cph.subdisthz.other,
  cov1 = cov1_new
)
head(predict.other)

# Create overall survival
pool.time = sort(unique(c(cph.subdisthz.prost$uftime, cph.subdisthz.other$uftime)))
cifs = data.frame(
  time = pool.time,
  surv = numeric(length(pool.time)),
  cif.prost = numeric(length(pool.time)),
  cif.other = numeric(length(pool.time))
)
```

```

)
for (i in 1:nrow(cifs)){
  if (!(i %in% predict.prost[,1])){
    if (cifs$time[i] < predict.prost[1,1])
      cifs$cif.prost[i] = 0
    else cifs$cif.prost[i] = cifs$cif.prost[i-1]
  }else cifs$cif.prost[i] = predict.prost[,2][predict.prost[,1] == cifs$time[i]]

  if (!(i %in% predict.other[,1])){
    if (cifs$time[i] < predict.other[1,1])
      cifs$cif.other[i] = 0
    else cifs$cif.other[i] = cifs$cif.other[i-1]
  }else cifs$cif.other[i] = predict.other[,2][predict.other[,1] == cifs$time[i]]
}
cifs$surv = 1-cifs$cif.prost-cifs$cif.other
head(cifs)

# Trajectory plot of CIFs and overall survival
plot(
  cifs$cif.prost ~ cifs$time,
  type="s", ylim=c(0,1), lwd=2, col="blue",
  xlab="Months from prostate cancer diagnosis",
  ylab='Probability',
)
lines(cifs$cif.other ~ cifs$time, type="s", col="green", lwd=2)
lines(cifs$surv ~ cifs$time, type="s", col="red", lwd=2)
legend(
  "topright",
  c(
    "Prostate",
    "Other",
    'Overall Survival'
  ),
  col=c('blue','green','red'), lwd=2
)

# Stacked plot of CIFs
cifs_long = tidyr::pivot_longer(
  cifs, cols = -time, names_to = "Types", values_to = "estimate")
library(ggplot2)
ggplot(data = cifs_long, aes(x = as.numeric(time), y = estimate, fill = Types)) +
  geom_area(alpha = 0.6) +
  labs(x = "Months from prostate cancer diagnosis", y = "Probability") +
  theme_minimal()

```